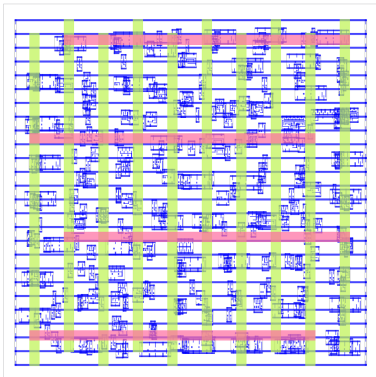


Four ways to do global placement

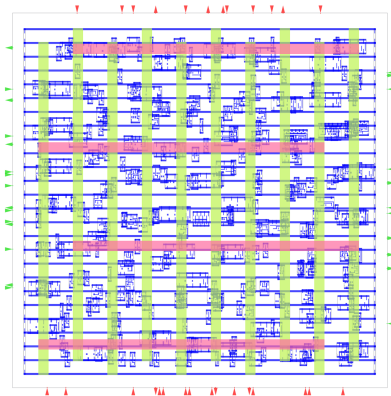
Mode	Pins	Needs
<code>global_placement</code>	Uses placed pins; <code>-skip_io</code> ignores them	Floorplan, cells, and rows
<code>-place_ios</code>	Co-optimizes movable pins and cells	Supported die and pin setup; run <code>place_pins</code> afterward
<code>-routability_driven</code>	Uses placed pins	Routing layers for congestion estimates; no prior routing or parasitic estimate
<code>-timing_driven</code>	Uses placed pins	Liberty, timing constraints, and wire RC; CTS is not required

Routing-aware placement setup



```
# Reserve part of the metal stack in  
%.  
# Use spare capacity for later  
routing.  
# Keep placement routable.  
# Set signal and clock layer ranges.  
set_global_routing_layer_adjustment \  
    metal2-metal10 0.5  
set_routing_layers \  
    -signal metal2-metal10 \  
    -clock metal6-metal10
```

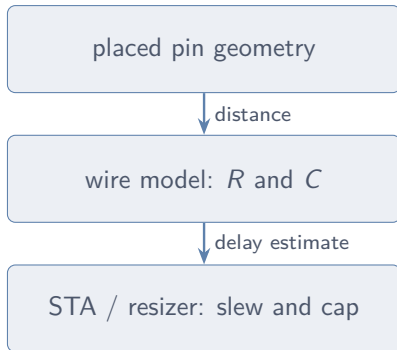
Final routability-driven placement



```
# Placement mode alternatives:
-place_ios
# Co-optimize pin placement with
# cell placement.
-skip_io
# Place cells without considering
# or needing pin locations.
# Density is between 0 and 1.
# Default is 0.7.

# Final pass: use congestion in the
# objective after pins are placed.
global_placement -routability_driven
\
    -density 0.80 -pad_left 2
    -pad_right 2
```

Estimate placement parasitics



placement-based parasitics
approximate future routed wires

```
# One set_layer_rc call per layer.  
# -layer takes one name, not a range.  
# Tcl foreach can reuse R/C values  
# across groups of layers.  
# E.g. M1-M4 and M6-M7, separately.  
# Override missing or unsuitable  
# LEF values only.  
set_layer_rc -layer metal3 \  
    -resistance 5e-3 \  
    -capacitance 5e-2  
  
# set_wire_rc chooses signal and  
# clock layers for wire estimates.  
set_wire_rc -signal -layer metal3  
set_wire_rc -clock -layer metal6  
# Use placed pin locations, not  
# detailed routed wires.  
estimate_parasitics -placement
```

repair_design: more information

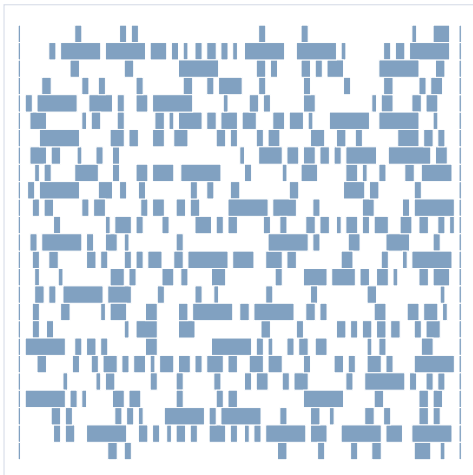
Check	What it limits	Information used
Maximum slew	How slowly a signal edge changes	Cell models and estimated wire RC
Maximum capacitance	Total load driven by a cell	Pin loads and estimated wire capacitance
Maximum fanout	Number of load pins on a net	Netlist connectivity

After placement, `repair_design` can repair all three. CTS is not required. It also buffers long wires.

Placement engines

- ▶ Supports mixed cell heights (1x–4x)
- ▶ Supports fence regions and fragmented rows
- ▶ Supports placement padding and filler insertion
- ▶ Two legalization engines:
 - ▶ **Negotiation legalizer (default):** handles dense placement, but is slower; uses diamond search for remaining corner cases.
 - ▶ **Diamond search (legacy):** faster search from each cell's starting position. Enable with `-use_diamond_legalizer`.

Detail placement



```
# Reserve space beside each cell.
set_placement_padding -global \
    -left 1 -right 1
# Move cells onto legal row sites.
# Resolve overlaps after repair.
detailed_placement
# Check for placement violations.
check_placement -verbose
```